

Phoneme → Text Decoder: Project Update

Homophone-aware P2T decoding for lip-reading on the LRS2 corpus
Llama-3.2-3B · QLoRA (4-bit NF4 + LoRA) · full-corpus run logged 27 June 2026 · brief prepared 8 July 2026

9.55%	5.33%	0.837	65.4%
Word Error Rate ↓	Char Error Rate ↓	BLEU-4 ↑	Exact Match ↑

A Llama-3.2-3B model was fine-tuned with QLoRA to decode ARPAbet phoneme strings into English sentences. On the 9,633-sentence evaluation split it decodes cleanly, and the difference between homophone and non-homophone sentences, the project's central question, is **small and inconsistent rather than large**.

Results (evaluation set, n = 9,633)

Subset	n	WER ↓	CER ↓	BLEU-4 ↑	Exact Match ↑
Overall	9,633	9.55%	5.33%	0.837	65.45%
Homophone	7,511	9.32%	5.32%	0.840	64.67%
Non-homophone	2,122	10.78%	5.40%	0.810	68.24%

A zero-shot baseline (an untuned small language model) scores about 280% WER, so the task is not solvable without fine-tuning.

Work completed

- A full P2T pipeline: grapheme-to-phoneme data prep, QLoRA training, and a stratified evaluation harness reporting WER, CER, BLEU-4, and exact match.
- Training on the full LRS2 corpus of 48,164 phoneme-sentence pairs, using a random 80/20 split (seed 42).
- A contrastive hard-negative objective on homophone examples, aimed at disambiguation.
- A zero-shot baseline to establish how hard the task is without fine-tuning.

Finding

- WER is slightly lower on homophone sentences (9.32 vs 10.78), while exact match is slightly lower on the non-homophone ones (64.67 vs 68.24).
- At this scale, homophone-heavy sentences are not much harder than the rest. The language model resolves most of the ambiguity from context.

Limitations

- No separate test set. Scores come from the 20% validation split, so they may overstate how well the model generalises.
- No ablation yet. Without a run that drops the contrastive loss, its effect on the homophone gap cannot be isolated.
- Hard-negative gap. The negatives place substituted word text in the phoneme slot instead of re-derived phonemes. The fix is identified.

Next steps

- Set aside a held-out test set for final numbers.
- Run the contrastive ablation, with and without the loss.
- Re-derive phonemes for substituted words, then retrain.
- Try a larger LoRA rank (r=48) and a longer schedule.

Source: RESULTS_SUMMARY.md, dryrun_checkpoints/overnight_llama32/, LRS2. Figures come from the validation split and should be read as directional until a held-out test set is scored.